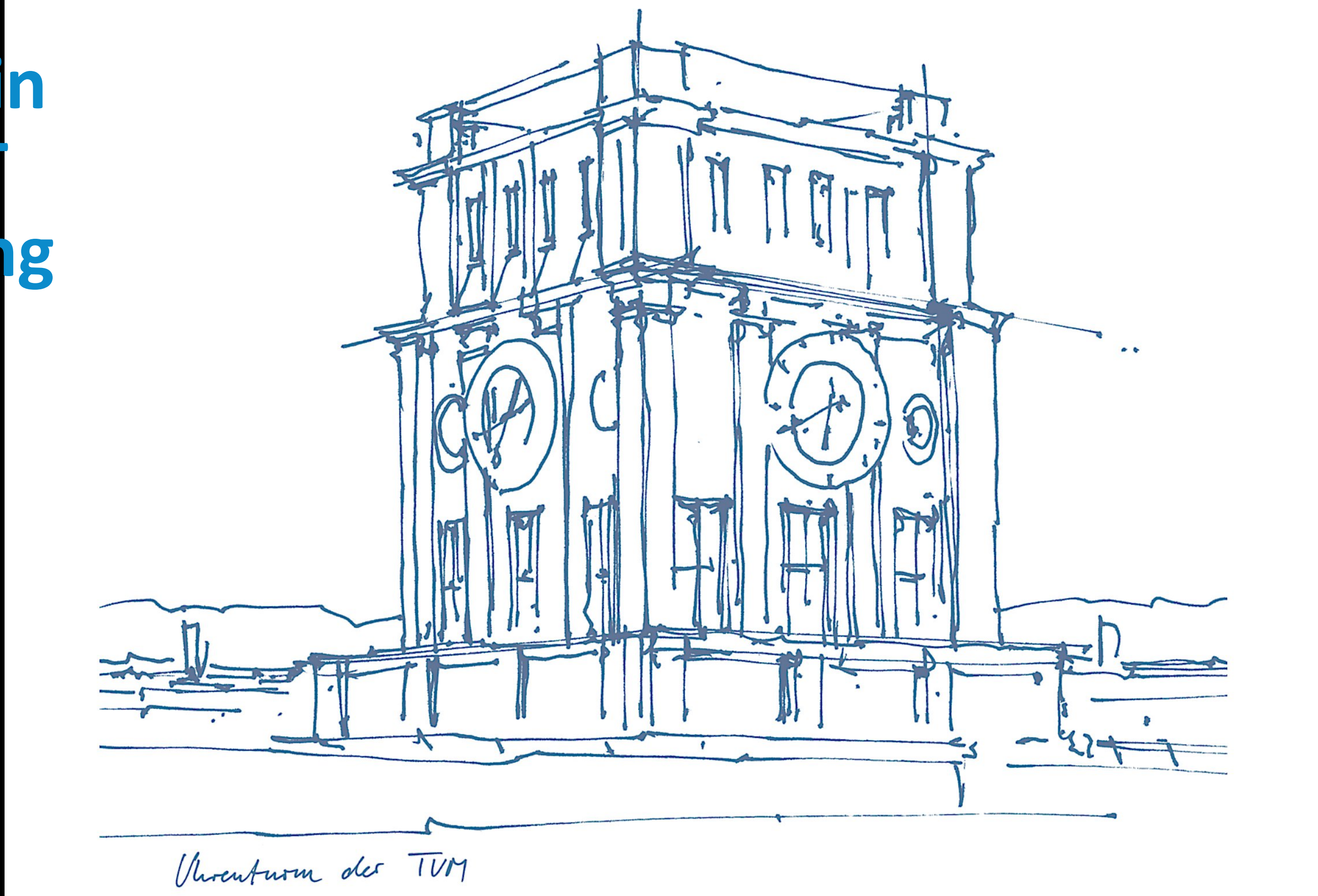
Streak Artifact Reduction in Human-scale Dark-field CT Using 3D Gaussian Splatting

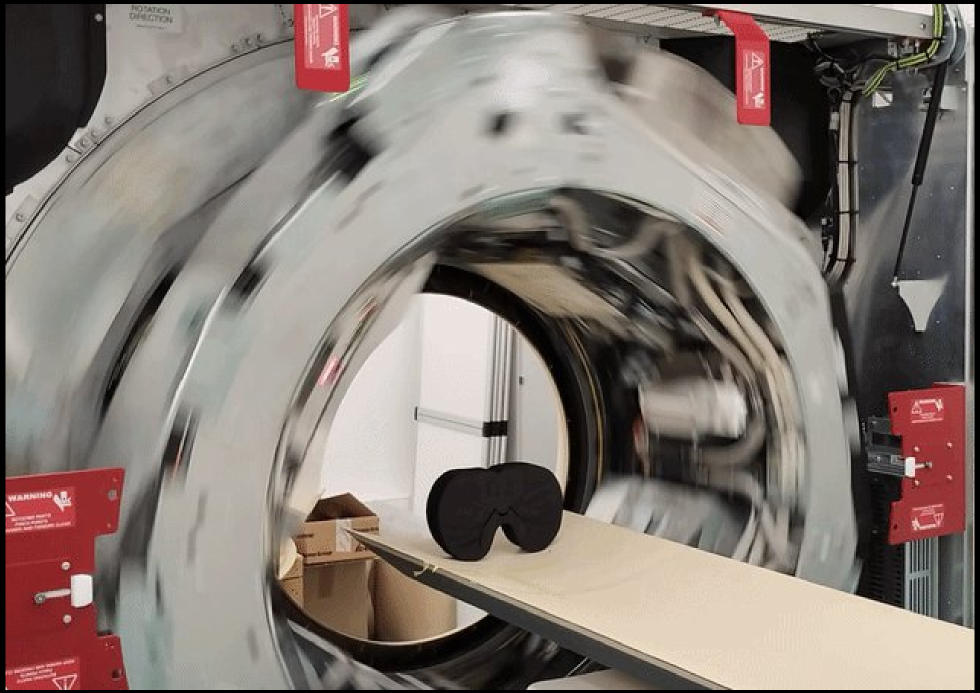


7th International Conference on X-ray and Neutron Phase Imaging with Gratings
Munich 24.06.2026

Tina Dorosti
Technical University of Munich
School of Natural Sciences
Department of Physics
Chair of Biomedical Physics

Motivation

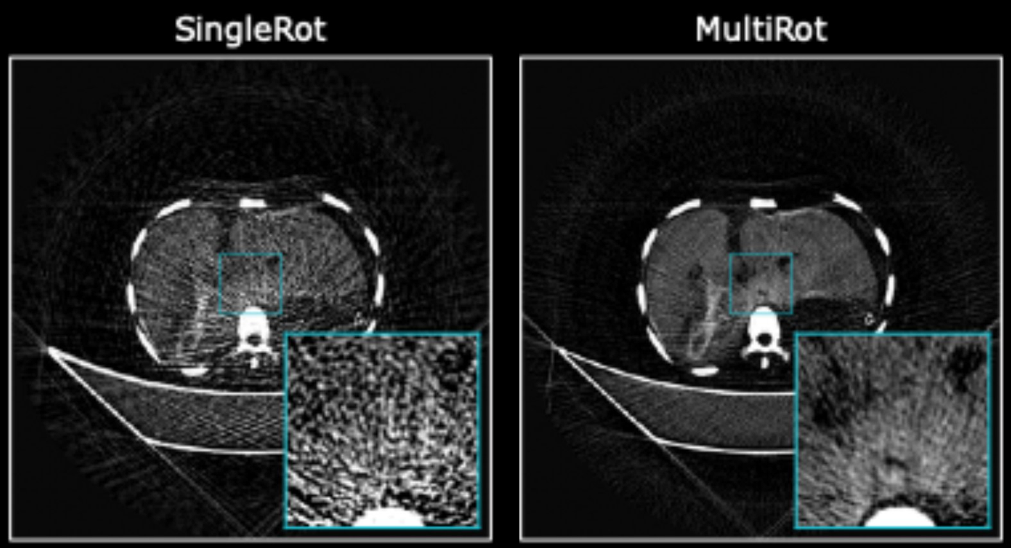
- Inherent streak artifacts due to the grating interferometry setup at the DFCT

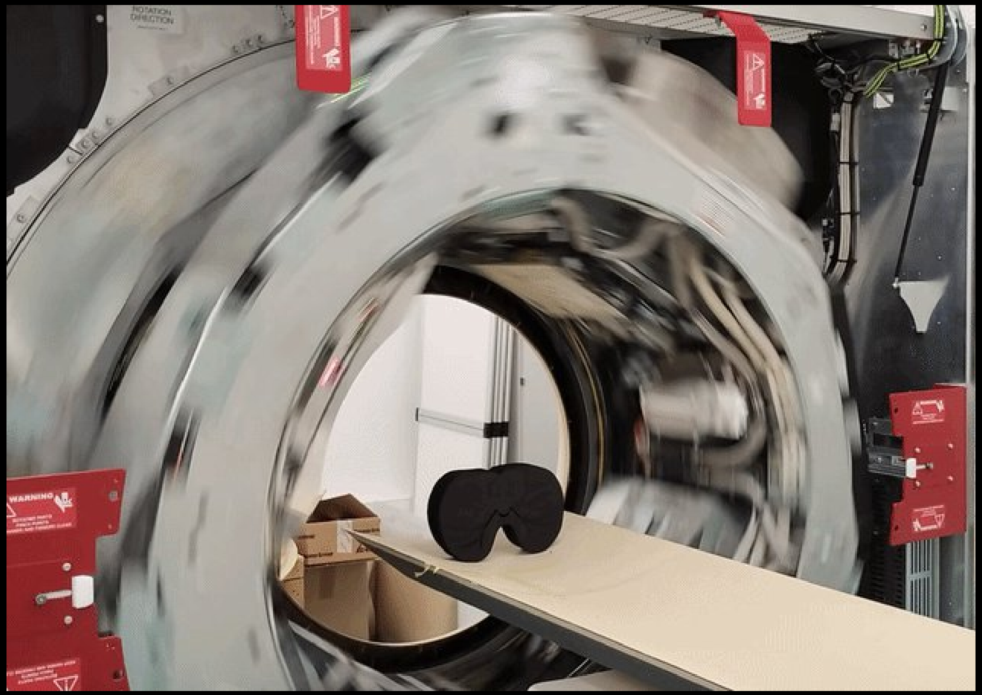


Taken by Daniel Frey

Motivation

- Inherent streak artifacts due to the grating interferometry setup at the DFCT
- More CT rotations result in cleaner images, but not clinically feasible
- Even with x10 or x20 rotations, some streaks are still visible

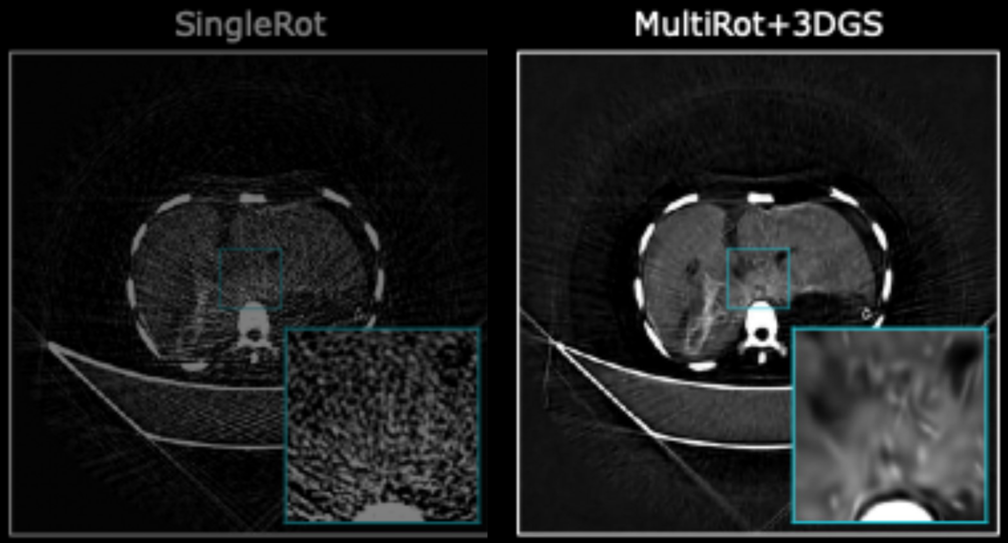




Taken by Daniel Frey

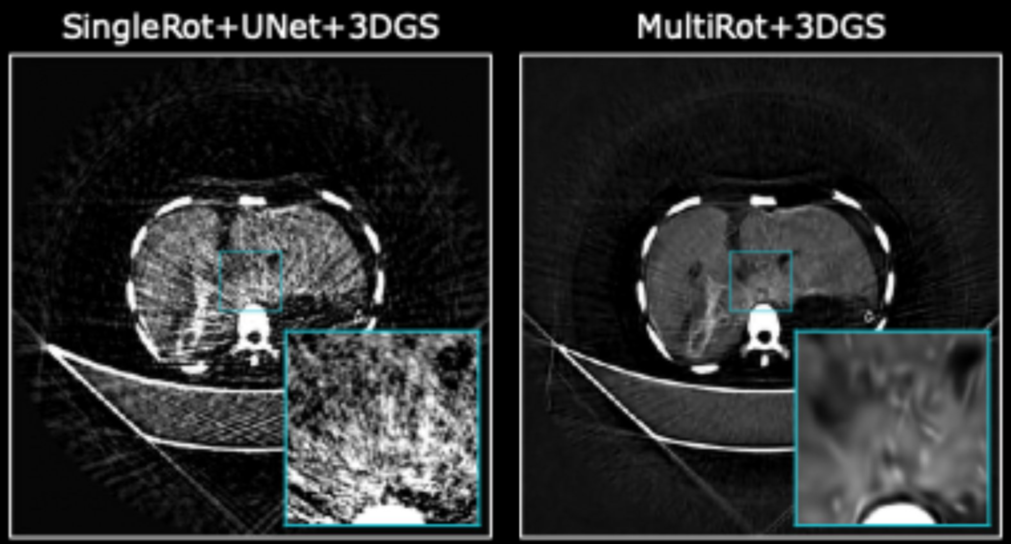
Idea

1. Improve the multi-rotation ‘ground truth’ reference



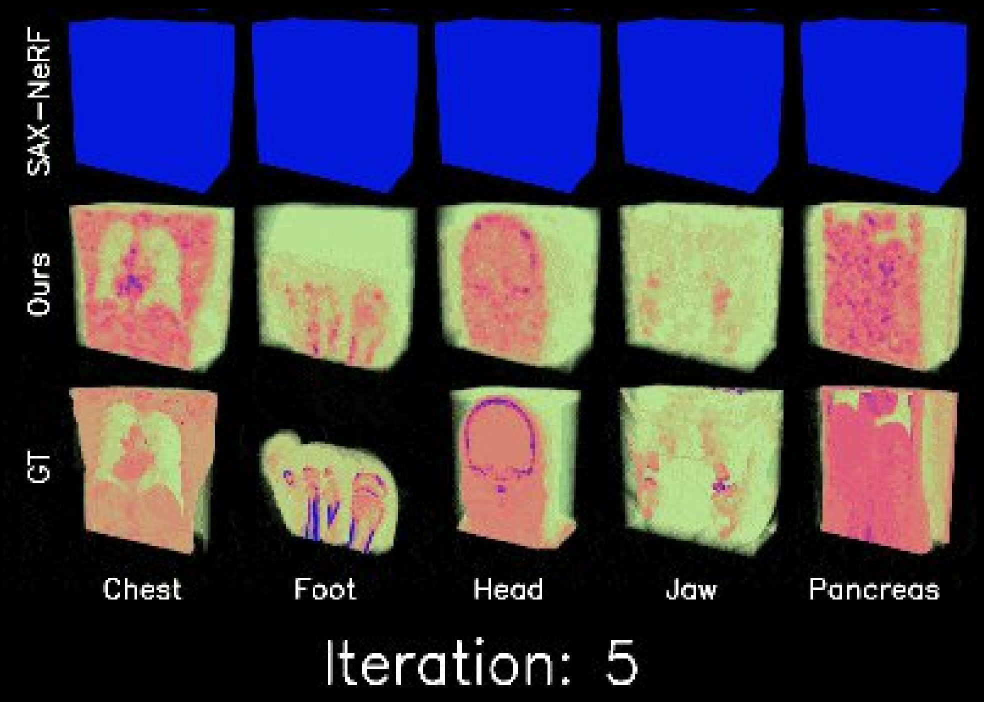
Idea

1. Improve the multi-rotation ‘ground truth’ reference
2. Benefit from this to improve the clinically relevant single-rotation data



Motivation

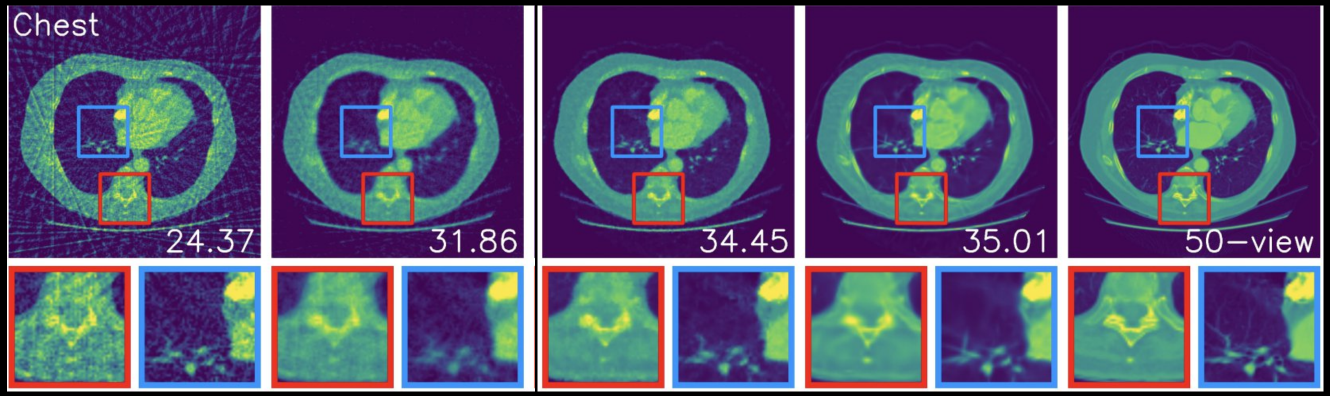
- Emerging methods with Neural Representation Field (NeRF) are promising but timely
- Gaussian splatting enables a faster approach to image rendering and space representation



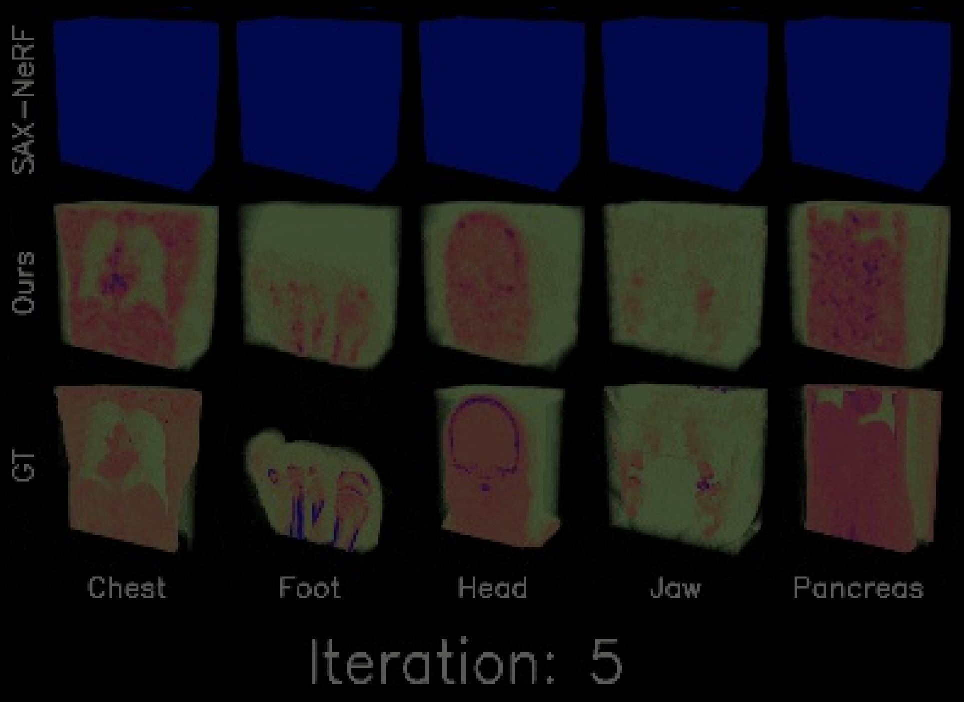
Adapted from R. Zha et al. (2024), doi: 10.48550/arXiv.2405.20693

Motivation

- Emerging methods with Neural Representation Field (NeRF) are promising but timely
- Gaussian splatting (3DGS) enables a faster approach to image rendering and space representation
- 3DGS improves sparse-view attenuation CT image quality



FDK SART SAX-NeRF Ours (R^2 3DGS) Ground Truth



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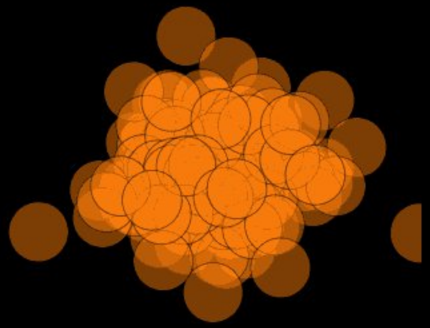
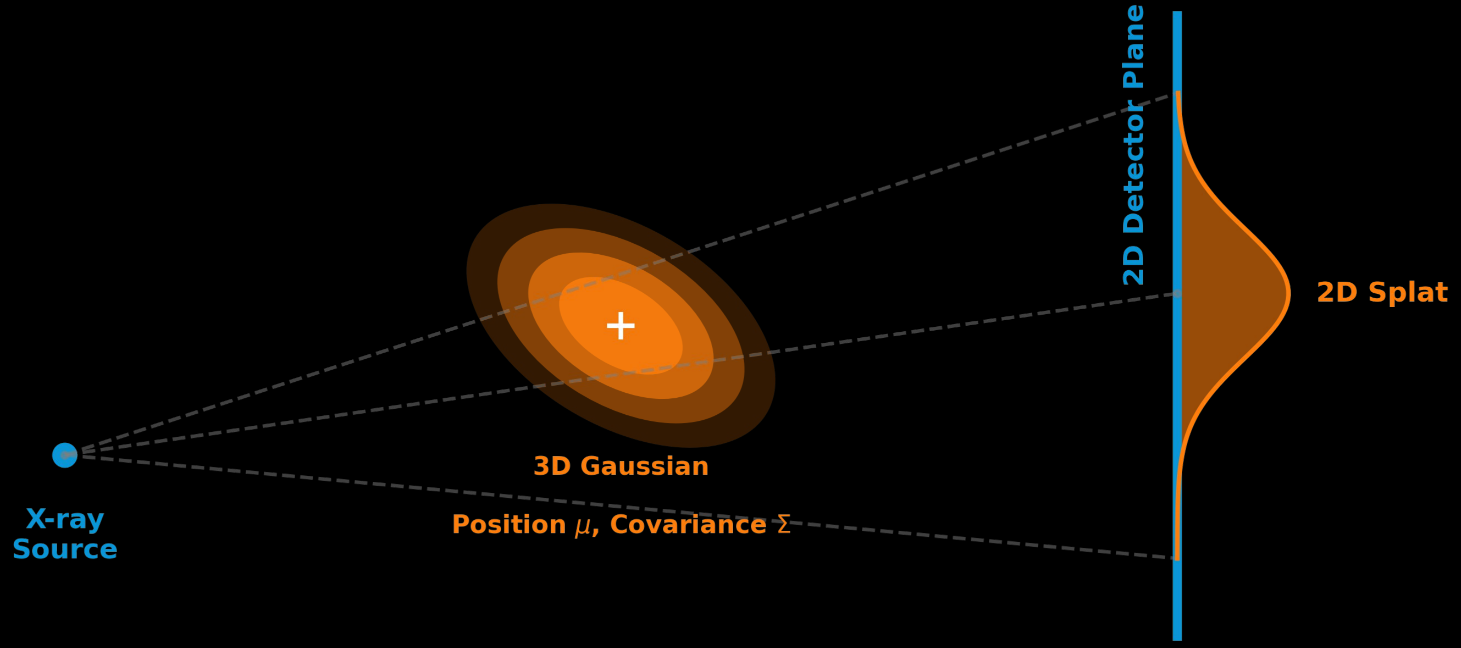
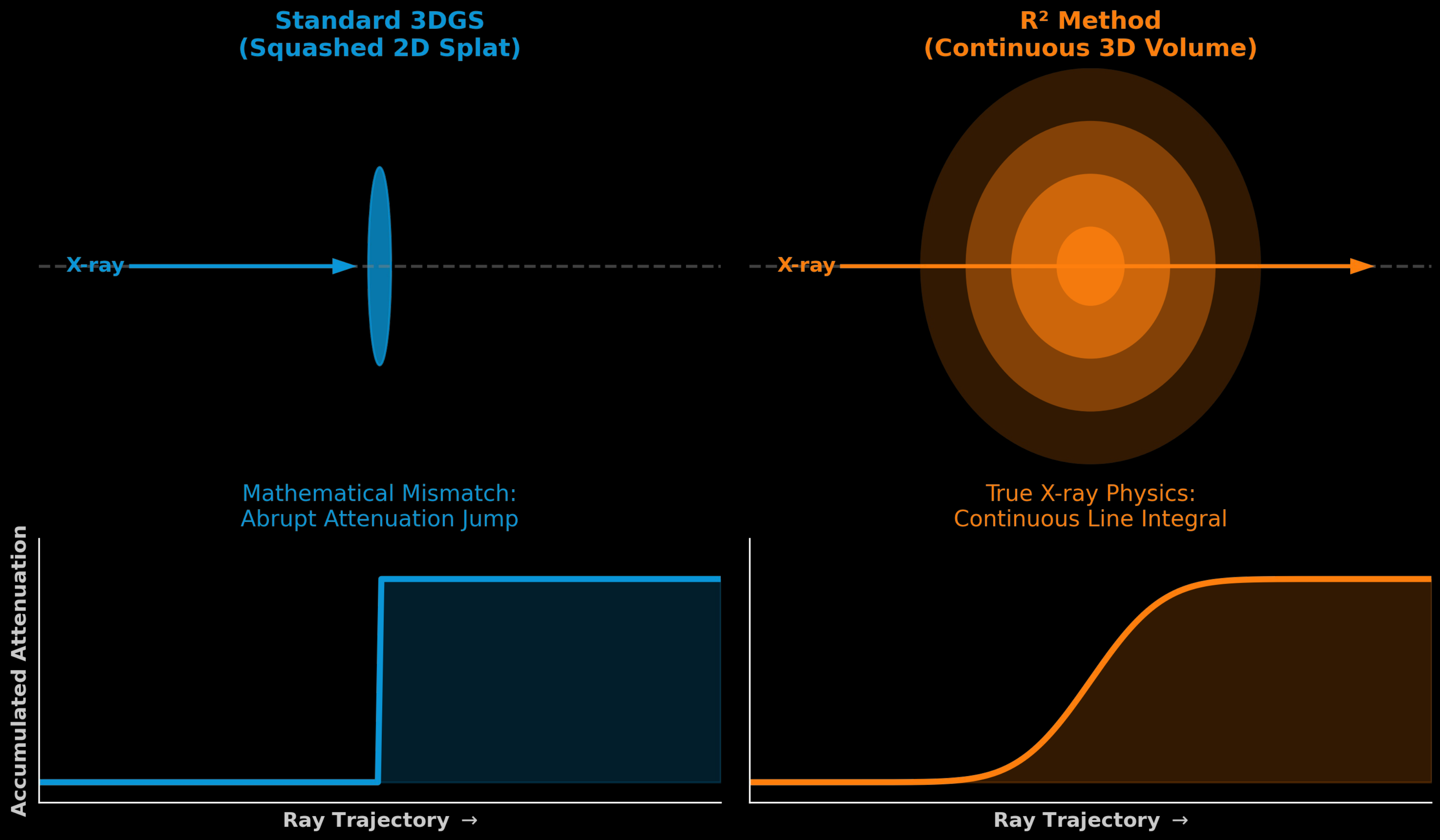
Gaussian Representation and Splatting

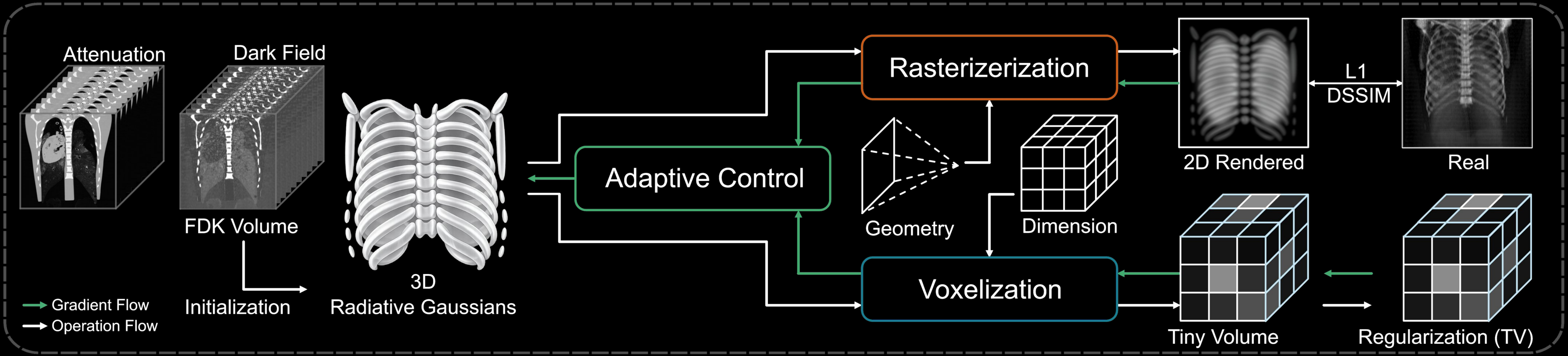




Rectified Radiative (R^2) Gaussian Splatting



Method: R^2 Gaussian Splatting



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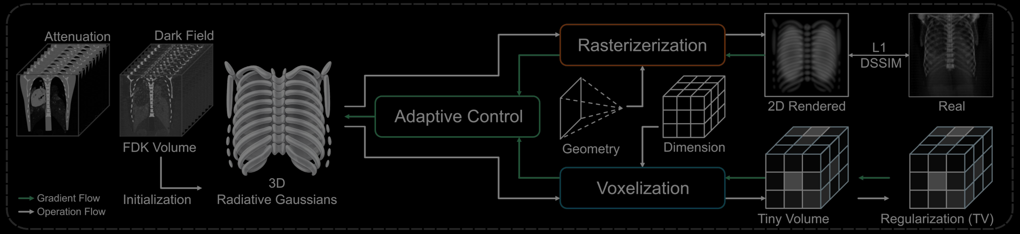
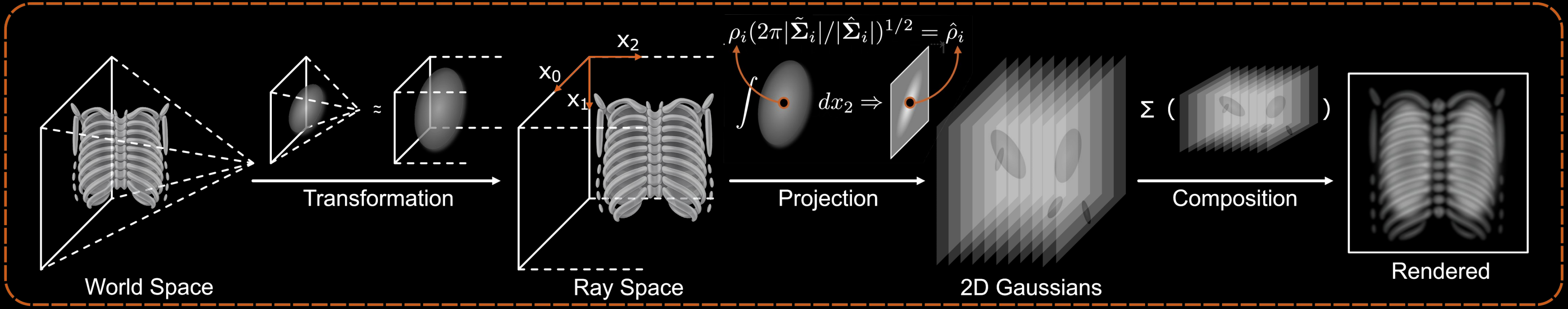
Gaussian Splatting





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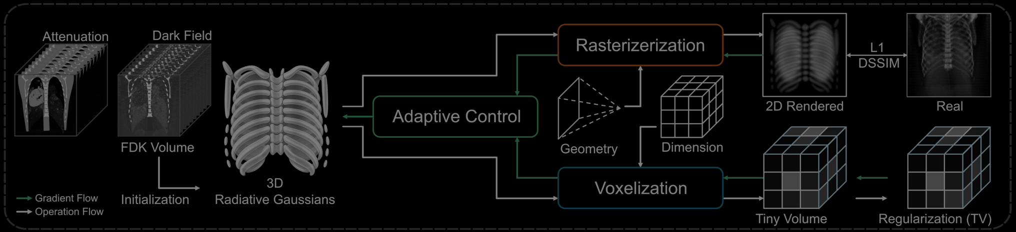

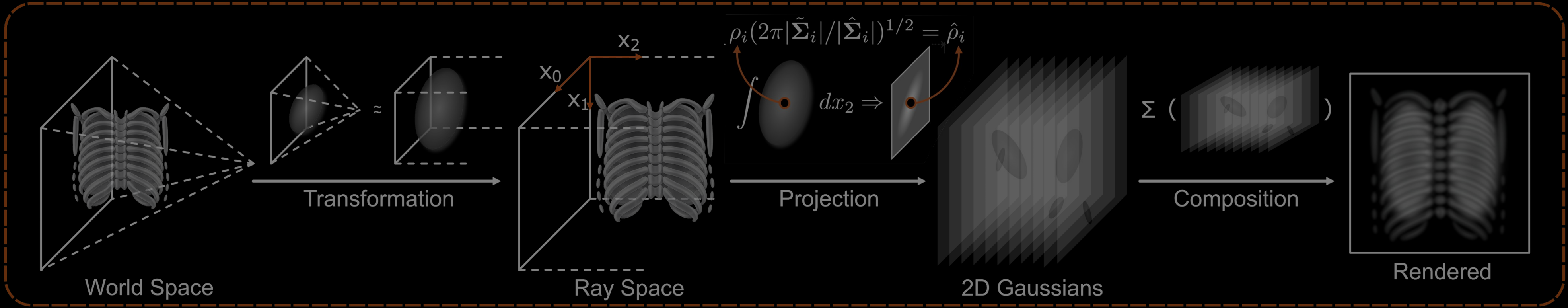

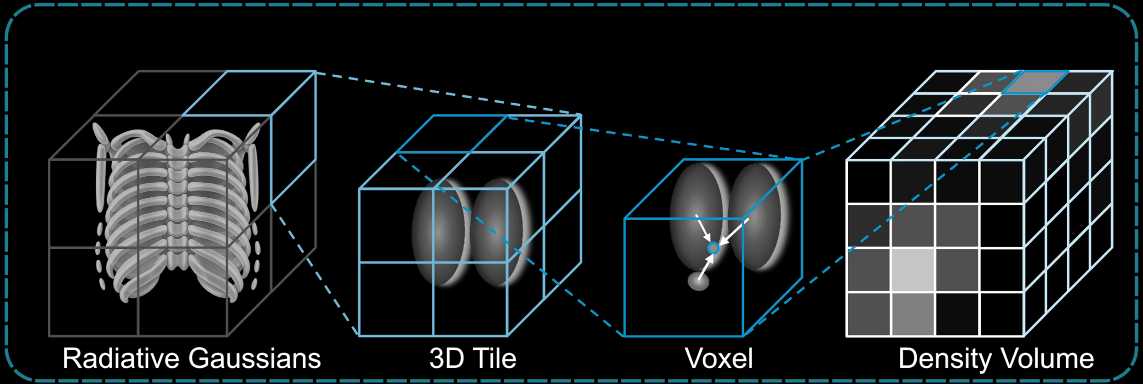

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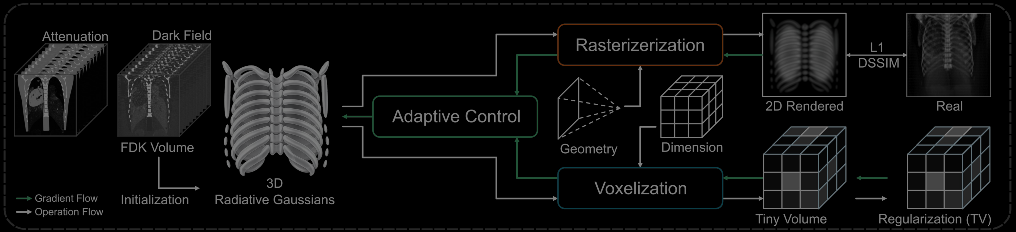

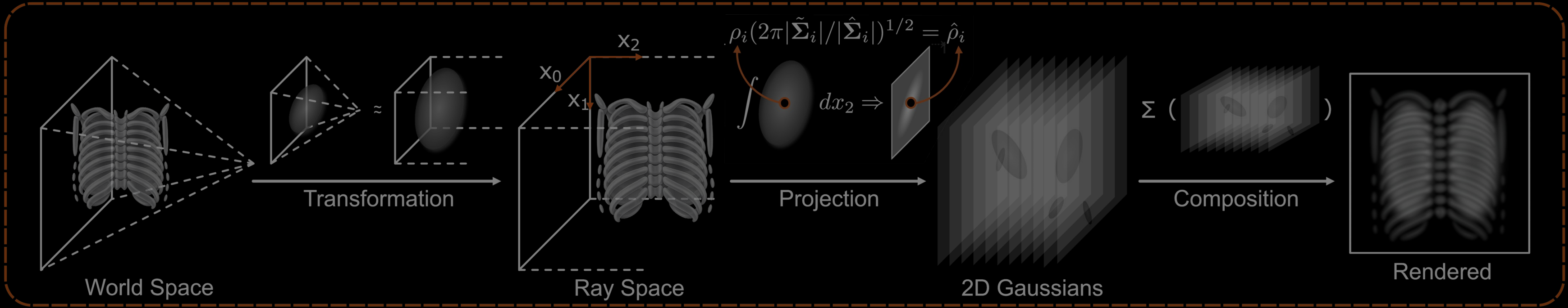

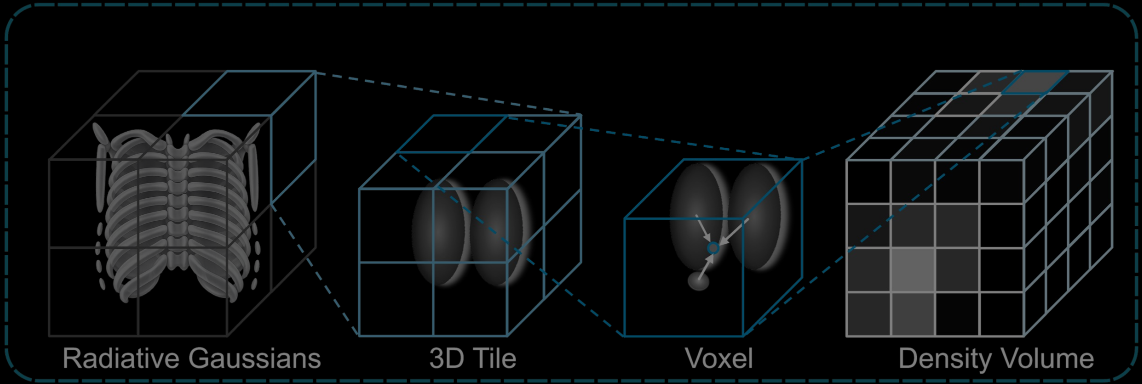

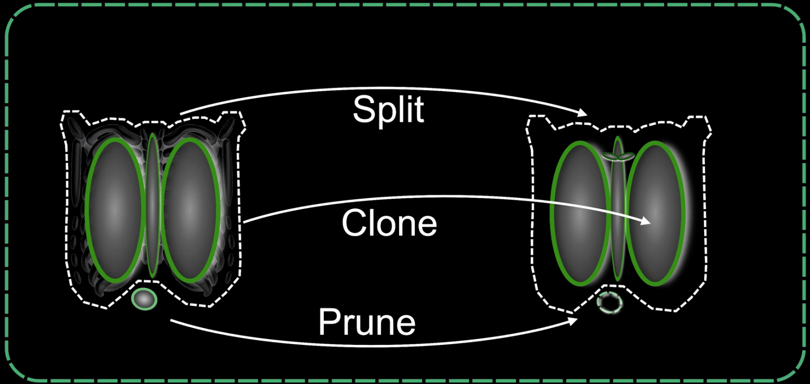

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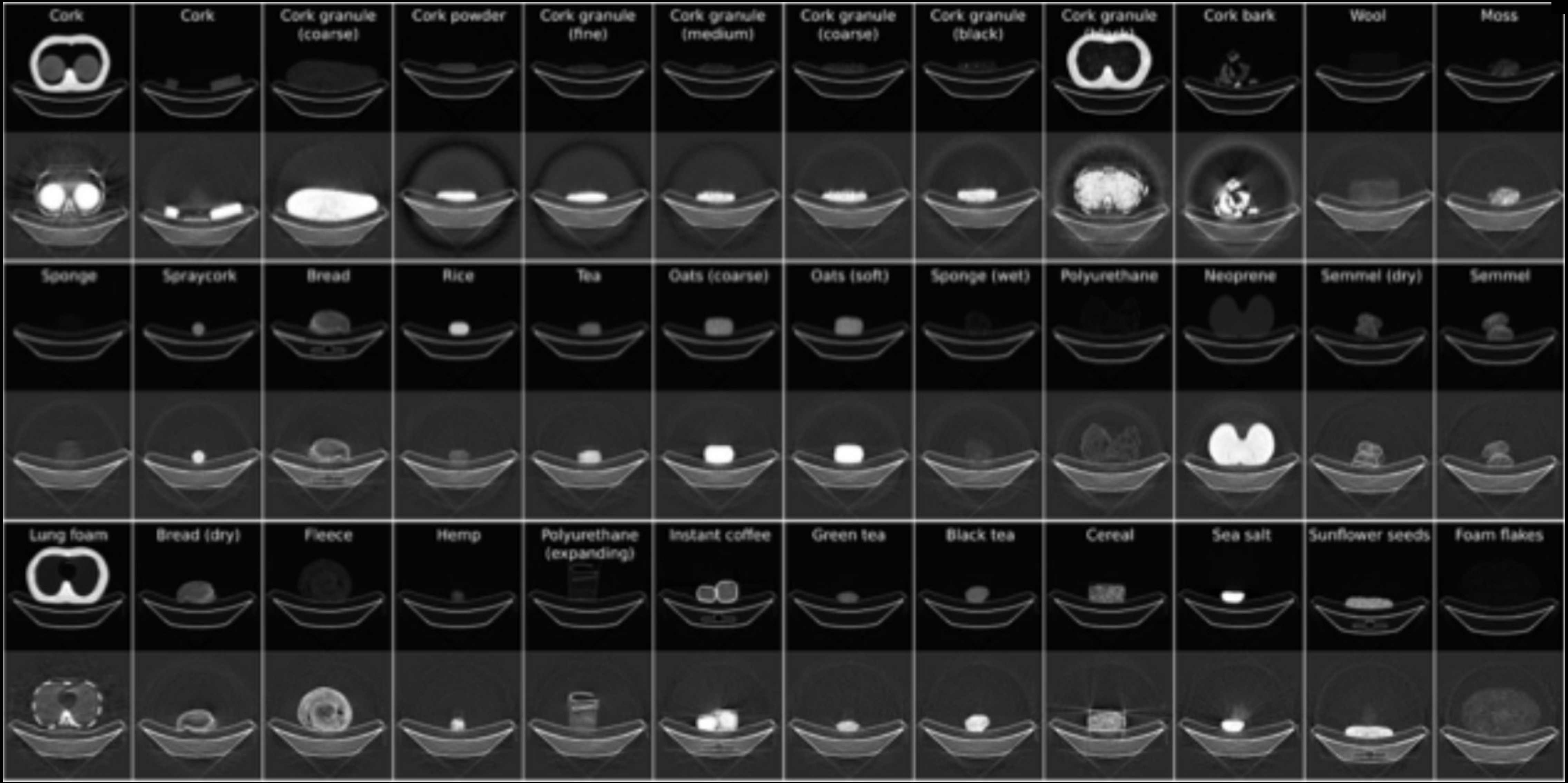
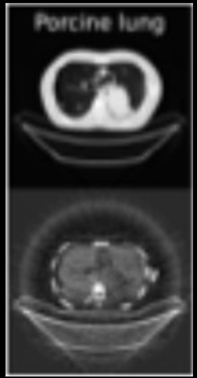
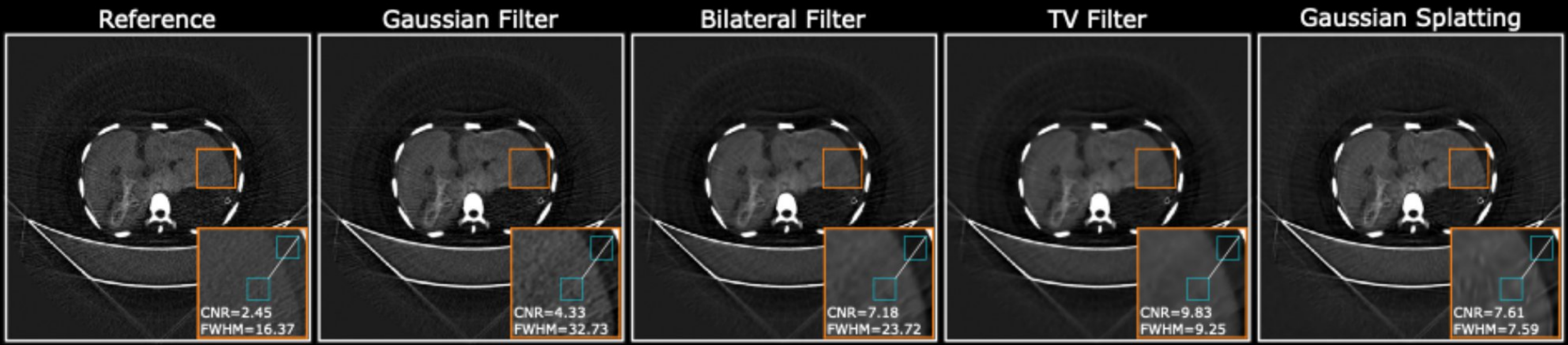
Data





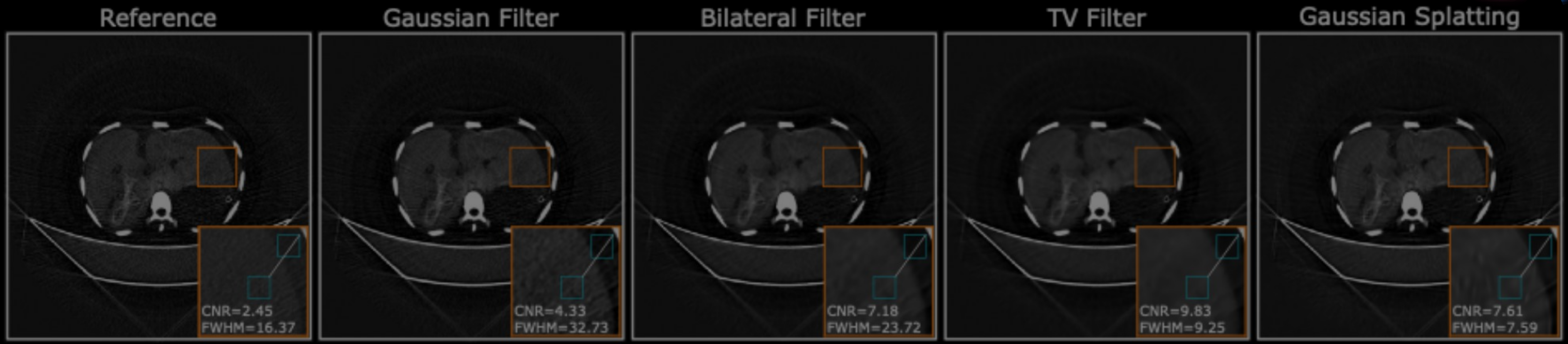
Daniel Frey – Master's Thesis (2025)

Results



Reference
Gaussian Filter
Bilateral Filter
TV Filter
Gaussian Splatting
CNR=2.45
FWHM=16.37
CNR=4.33
FWHM=32.73
CNR=7.18
FWHM=23.72
CNR=9.83
FWHM=9.25
CNR=7.61
FWHM=7.59

Results



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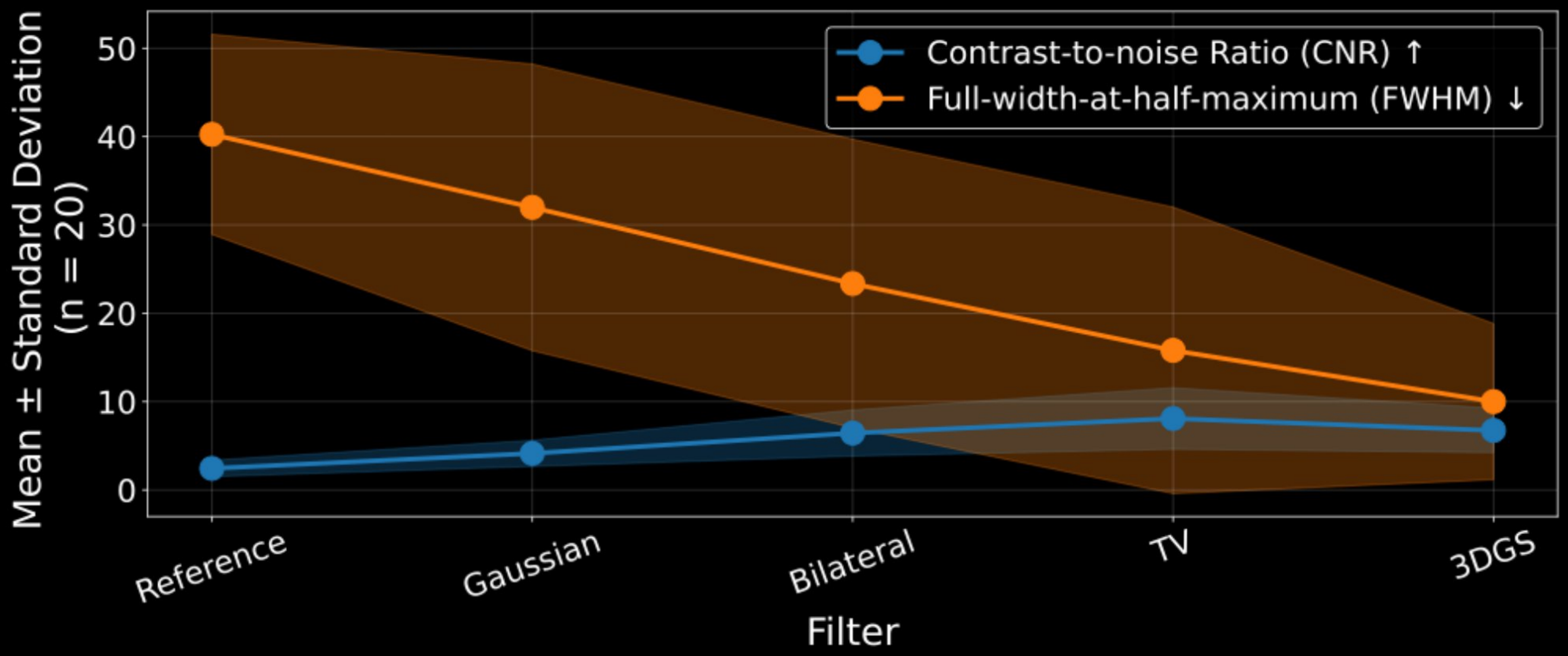
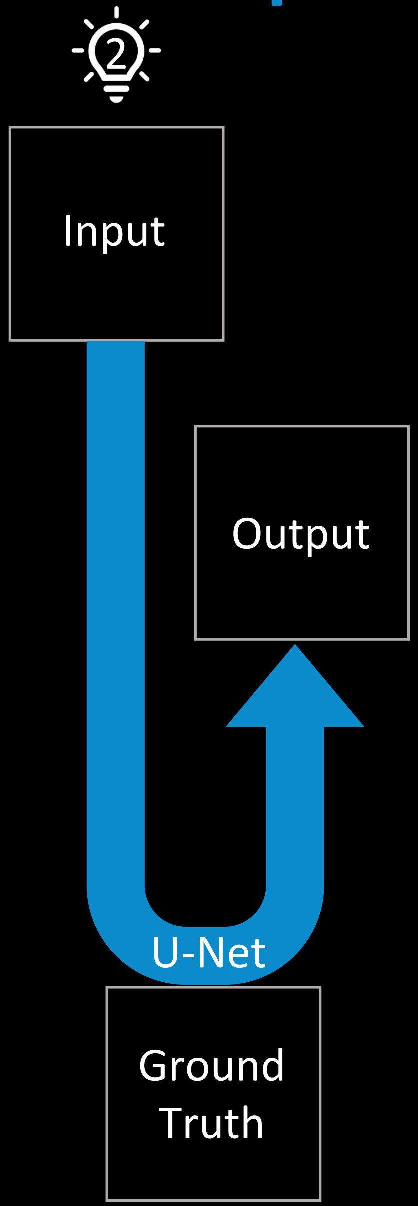


Contrast-to-noise Ratio (CNR) ↑
Full-width-at-half-maximum (FWHM) ↓
Mean ± Standard Deviation (n = 20)
0
10
20
30
40
50
Reference
Gaussian
Bilateral
TV
3DGS
Filter

Next Step: Supervised Learning



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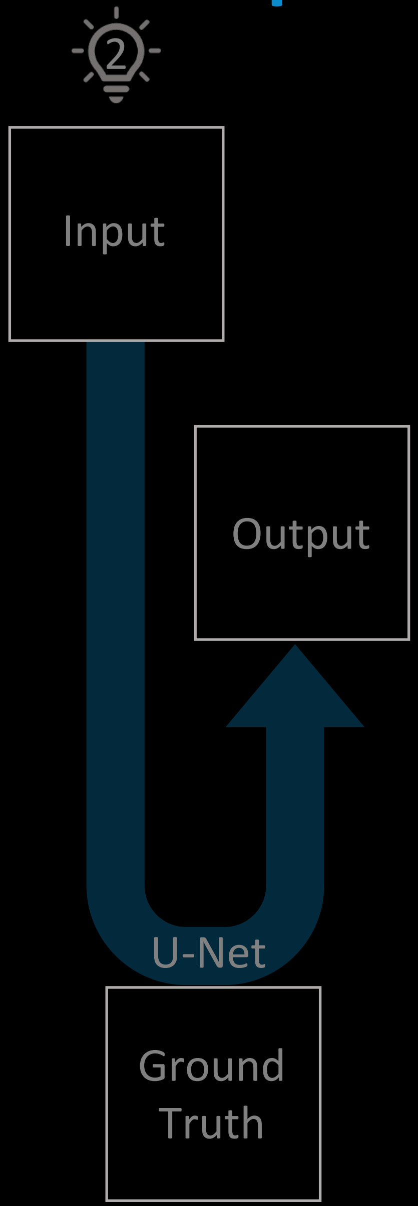

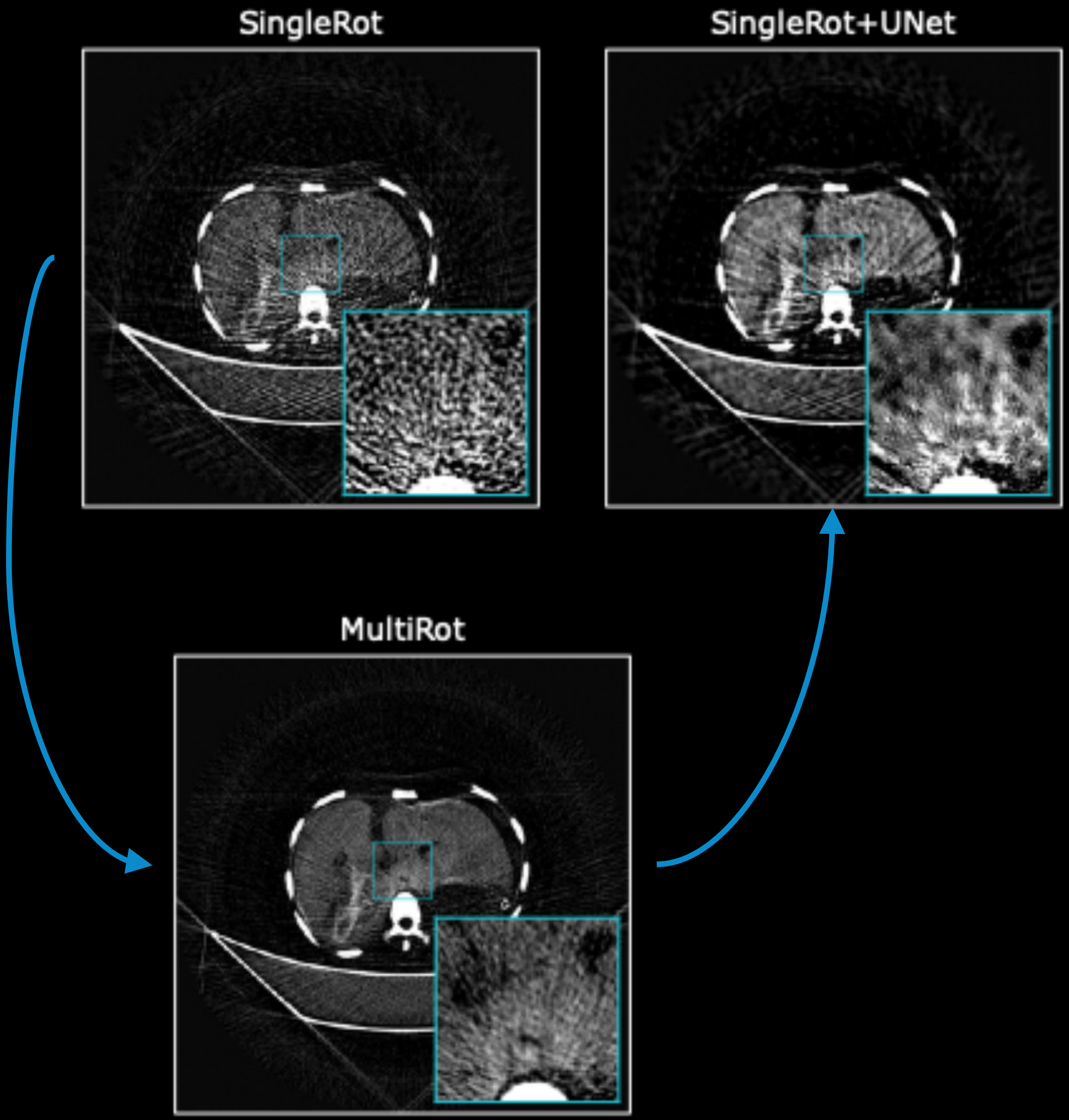

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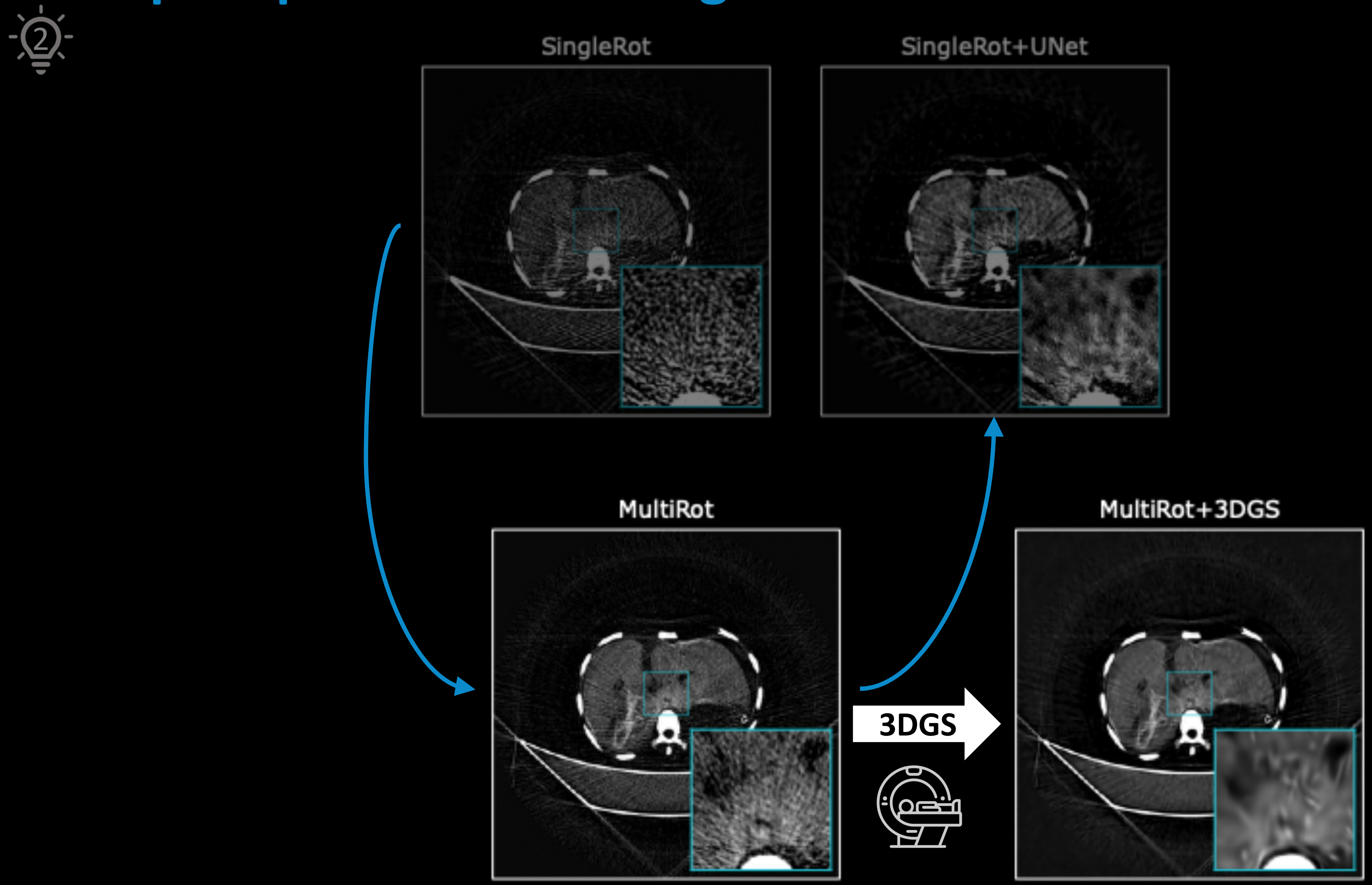

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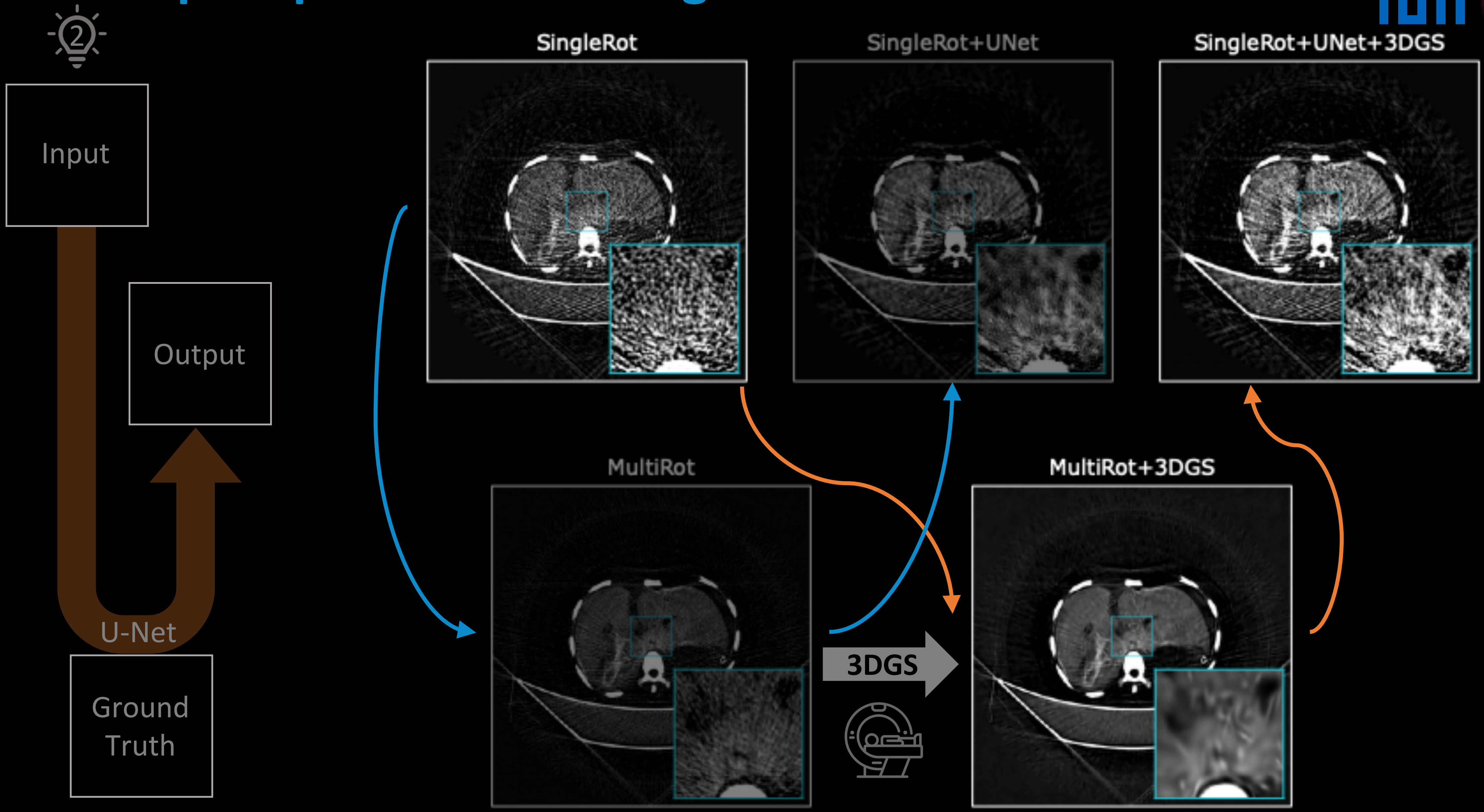
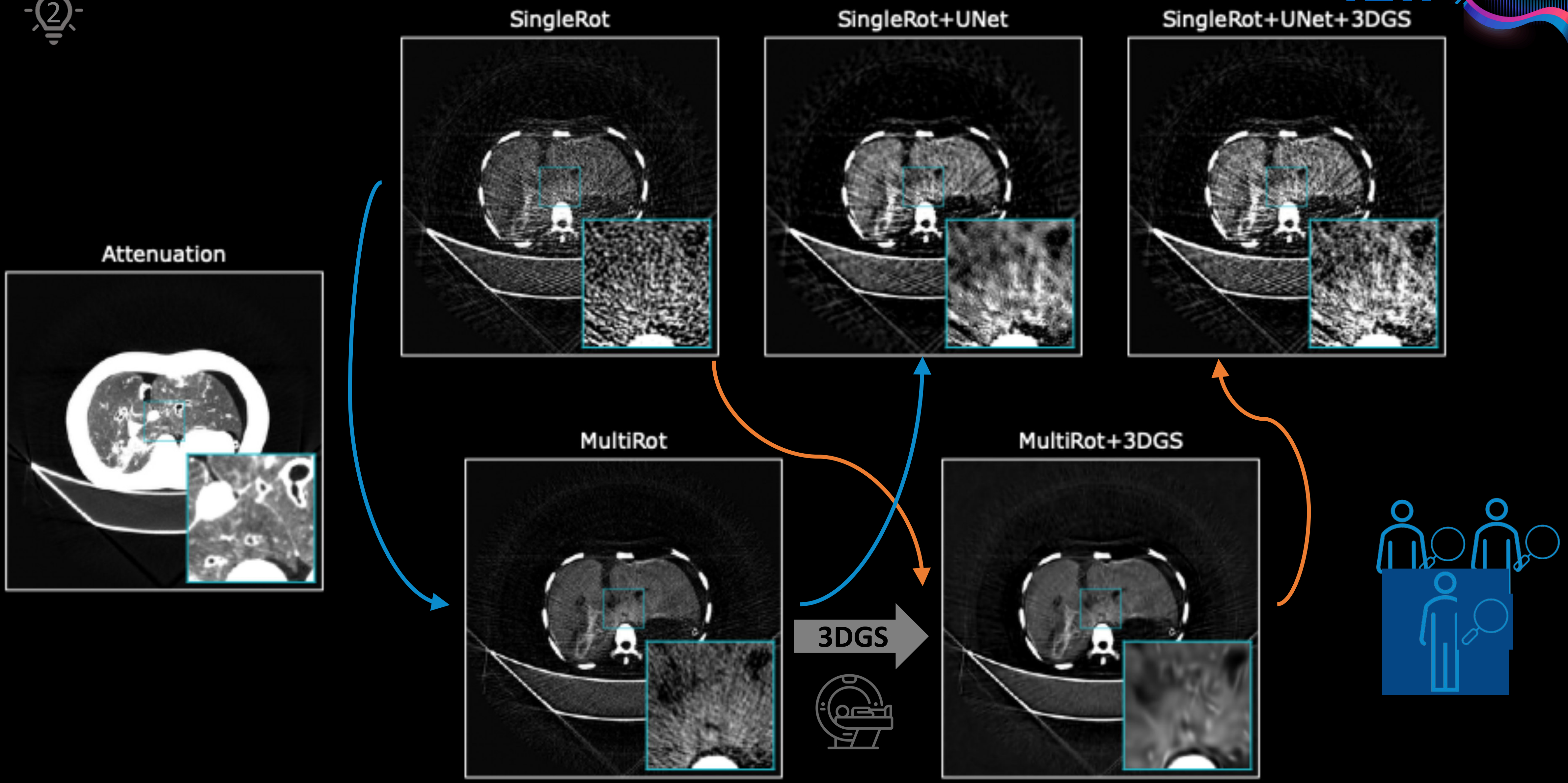


Next Step: Reader Evaluation



Thank You!

DFCT suffers from inherent streak artifacts

3DGS outperforms conventional filtering in DFCT streak-reduction

Necessities for future clinical trials: Reduced rotation and improved image quality,

Bayerisches Staatsministerium für Wissenschaft und Kunst

Federal Ministry of Education and Research